Modelling a 3D Printer

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0 Introduction

3D printers have the basic task of extruding filament at a given x, y, z coordinate at time t. The mechanism that achieves that movement determines the machine's costs, print quality, and robustness. Out of many different options, the Core XY mechanism is a popular choice for modern printers. Unlike Cartesian designs, where one motor drives one axis, here each motor simultaneously changes the x and y position of the tool head (see Figure 1). This results in a more complex kinematic model, but allows a sturdier casing and faster printing speeds. Our task is to model and simulate the system.

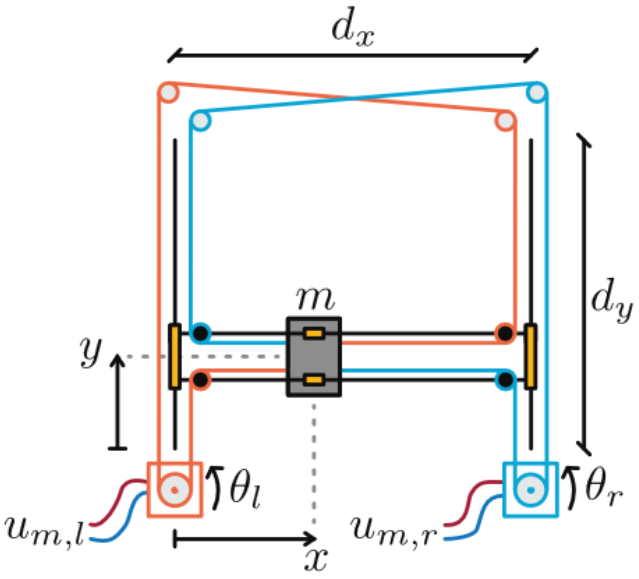


Figure 1: Schematic of the Core XY mechanism.

0.1 Symbol Table

Table 1: Symbols and parameter values.

Description	Symbol	Value
Motor angle left	θ_L	–
Motor voltage left	$u_{m,L}$	–
Motor current left	i_L	–
Motor radius	r_m	11 mm
Motor inductance	L_m	10 mH
Motor resistance	R_m	90 Ω
Motor inertia	J_m	8×10^{-6} kg m^2
Motor constant	K_m	0.07 N m/A
Tool head position	x, y	–
Tool head mass	m	60 g
Nozzle voltage	u_n	–
Nozzle inductor current	$i_{L,n}$	–
Nozzle capacitance	C_n	–
Nozzle inductance	L_n	–
Capacitor voltage	u_c	–
Nozzle thermal capacity	$C_{t,n}$	–
Nozzle thermal resistance	$R_{t,n}$	9 K/W
Nozzle resistance	R_n	8 Ω
Resistor thermal capacity	C_r	–
Resistor temperature	T_r	–
Thermal resistance to environment	R_{env}	–
Reference resistance	R_0	–
Reference temperature	T_0	–
Temperature coefficient of resistance	α	–
Nozzle temperature	T_n	240 °C
Environmental temperature	T_e	26.85 °C
Bearing friction rolling	μ_r	2.7×10^{-8} N m s/rad
Bearing friction linear x-direction	$\mu_{l,x}$	0.002 N s/m
Bearing friction linear y-direction	$\mu_{l,y}$	0.001 N s/m
Bearing radius	r_b	3 mm
Print bed temperature	T_b	105 °C
Print area dimension x	d_x	29 cm
Print area dimension y	d_y	29 cm

1 Physical Modelling

1.1 Kinematics

The first task is to derive the direct kinematics of the Core XY mechanism. This is slightly complicated by the fact that rotating one motor will move the tool head in both x and y directions. By removing the right motor, we can see that by moving the tool head in the positive x direction **or** in the positive y direction, the left motor will rotate in the positive direction. We can draw similar conclusions for the right motor, for which we get the following equations:

$$\Delta L = \Delta x + \Delta y, \qquad \Delta R = \Delta x - \Delta y, \qquad \Delta x = \frac{\Delta L + \Delta R}{2}, \qquad \Delta y = \frac{\Delta L - \Delta R}{2} \tag{1.1}$$

With ΔL and ΔR being the length of belt moved by the motors, and Δx and Δy being the change in position of the tool head.

$$\Delta L = r_m \, \theta_L, \qquad \Delta R = r_m \, \theta_R \tag{1.2}$$

If we assume that in the beginning the motors are at $\theta_L = \theta_R = 0$ and the tool head is at $(0, 0)$, we get the following equations:

$$x(\theta_L, \theta_R) = \frac{r_m\theta_L + r_m\theta_R}{2} = \frac{r_m}{2}\left(\theta_L + \theta_R\right), \qquad y(\theta_L, \theta_R) = \frac{r_m\theta_L - r_m\theta_R}{2} = \frac{r_m}{2}\left(\theta_L - \theta_R\right) \tag{1.3}$$

$$\dot{x}(\dot{\theta}_L, \dot{\theta}_R) = \frac{r_m}{2}\left(\dot{\theta}_L + \dot{\theta}_R\right), \qquad \dot{y}(\dot{\theta}_L, \dot{\theta}_R) = \frac{r_m}{2}\left(\dot{\theta}_L - \dot{\theta}_R\right) \tag{1.4}$$

1.2 Dynamics

In the next part we model the dynamics of the 3D printer system by applying the Euler–Lagrange equation. Additionally, we will split the problem into a mechanical and an electrical one, with the angular velocity of the motor shafts connecting them.

Because of that, we choose generalised coordinates:

$$q = \begin{bmatrix} x \\ y \\ q_L \\ q_R \end{bmatrix} \qquad \dot{q} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ i_L \\ i_R \end{bmatrix} \tag{1.5}$$

We can also then split them among our two problems:

$$\dot{q}_m = \begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} \text{ (mechanical part)}, \qquad \dot{q}_e = \begin{bmatrix} i_L \\ i_R \end{bmatrix} \text{ (electrical part)} \tag{1.6}$$

1.2.1 Mechanical Part

Kinetic Energy

Firstly, we derive the kinetic energy terms. In the *mechanical* part, the only mass is the tool head. With that we have:

$$K_{\text{mech}} = \tfrac{1}{2} m(\dot{x}^2 + \dot{y}^2) \tag{1.7}$$

As the whole system is planar, we can see that there are no potential energy terms as it will be constant.

Dissipative Energy

Secondly, we find the dissipative energy terms. Each motor has 4 rotational bearings, and 4 linear bearings for the whole system, two along each axis. We define r_b as the bearing radius, later it will be kept the same as the motor radius for simplicity. From (1.3), we can get transformations (1.8) with trivial time derivatives.

$$\theta_{b,L1} = \frac{x+y}{r_b}, \qquad \theta_{b,R1} = \frac{x-y}{r_b} \tag{1.8}$$

These are used for the two fixed bearings on each belt. Then, for each side we get rotational dissipative energy:

$$D_{L1} = 2 \cdot \tfrac{1}{2} \cdot \dot{\theta}_{b,L1}^2 \, \mu_r = \mu_r \frac{1}{r_b^2} \left[\dot{x}^2 + 2\dot{x}\dot{y} + \dot{y}^2\right]$$
$$D_{R1} = 2 \cdot \tfrac{1}{2} \cdot \dot{\theta}_{b,R1}^2 \, \mu_r = \mu_r \frac{1}{r_b^2} \left[\dot{x}^2 - 2\dot{x}\dot{y} + \dot{y}^2\right]$$

For the two carriage bearings, if we assume the belt to be horizontal on the carriage like Figure 1 shows them, they are rotating only when the carriage moves along the x axis.

$$\theta_{b,L2} = \frac{x}{r_b}, \qquad \theta_{b,R2} = \frac{x}{r_b} \tag{1.9}$$

With that we get two more terms:

$$D_{L2} = 2 \cdot \tfrac{1}{2} \cdot \dot{\theta}_{b,L2}^2 \, \mu_r = \mu_r \frac{1}{r_b^2} \dot{x}^2, \qquad D_{R2} = 2 \cdot \tfrac{1}{2} \cdot \dot{\theta}_{b,R2}^2 \, \mu_r = \mu_r \frac{1}{r_b^2} \dot{x}^2$$

Then simply combining them:

$$\boxed{D_{\text{rot}} = \frac{\mu_r}{r_b^2} \left[4\dot{x}^2 + 2\dot{y}^2\right]}$$

Similarly, we get the linear dissipative energy:

$$D_{\text{lin}} = 2 \cdot \tfrac{1}{2} \mu_{l,x} \, \dot{x}^2 + 2 \cdot \tfrac{1}{2} \mu_{l,y} \, \dot{y}^2 = \mu_{l,x} \, \dot{x}^2 + \mu_{l,y} \, \dot{y}^2$$

Then, combining the terms we get:

$$D_{\text{mech}} = \left[4 \frac{\mu_r}{r_b^2} + \mu_{l,x}\right] \dot{x}^2 + \left[2 \frac{\mu_r}{r_b^2} + \mu_{l,y}\right] \dot{y}^2 \tag{1.10}$$

1.2.2 Electrical Part

Now we model the energies of the electrical part of the system. Keep in mind that both motors are the same.

Kinetic Energy

The inductor and the inertial element will contribute to the kinetic energy of this part of the system. Similarly to (1.8), using the motor shaft radius:

$$K_L = \tfrac{1}{2} i_L^2 L_m + \tfrac{1}{2} J_m \dot{\theta}_L^2 = \tfrac{1}{2} i_L^2 L_m + \tfrac{1}{2} J_m \frac{1}{r_m^2} \left(\dot{x}^2 + \dot{y}^2 + 2\dot{x}\dot{y}\right)$$
$$K_R = \tfrac{1}{2} i_R^2 L_m + \tfrac{1}{2} J_m \frac{1}{r_m^2} \left(\dot{x}^2 + \dot{y}^2 - 2\dot{x}\dot{y}\right)$$

Combining them we get:

$$K_{\text{el}} = \tfrac{1}{2} L_m \left[i_R^2 + i_L^2 \right] + J_m \frac{1}{r_m^2} \left(\dot{x}^2 + \dot{y}^2 \right) \tag{1.11}$$

As there are no capacitors in the system, there is no potential energy change as in the mechanical part.

Dissipative Energy

The only dissipative term in the electrical part of the system is the resistor:

$$D_{\text{el},L} = \tfrac{1}{2} i_L^2 R_m, \qquad D_{\text{el},R} = \tfrac{1}{2} i_R^2 R_m, \qquad D_{\text{el}} = \tfrac{1}{2} R_m \left(i_L^2 + i_R^2 \right) \tag{1.12}$$

Matrix Equations

Finally, we realise that the system is linear, and write the matrix expressions:

$$K = \tfrac{1}{2}\, \dot{q}^\top \mathbf{M}\, \dot{q}, \qquad D = \tfrac{1}{2}\, \dot{q}^\top \mathbf{N}\, \dot{q} \tag{1.13}$$

With corresponding matrices:

$$\mathbf{M} = \begin{bmatrix} m + J_m \frac{2}{r_m^2} & 0 & 0 & 0 \\ 0 & m + J_m \frac{2}{r_m^2} & 0 & 0 \\ 0 & 0 & L_m & 0 \\ 0 & 0 & 0 & L_m \end{bmatrix}, \qquad \mathbf{N} = \begin{bmatrix} 8 \frac{\mu_r}{r_b^2} + 2\mu_{l,x} & 0 & 0 & 0 \\ 0 & 4 \frac{\mu_r}{r_b^2} + 2\mu_{l,y} & 0 & 0 \\ 0 & 0 & R_m & 0 \\ 0 & 0 & 0 & R_m \end{bmatrix} \tag{1.14}$$

1.2.3 Generalised Force

Mechanical Coordinates

The non-conservative forces in the mechanical part are given by the motor torques τ_L and τ_R. The virtual work they perform is:

$$\Delta W = \tau_L \, \Delta\theta_L + \tau_R \, \Delta\theta_R$$

With the familiar transformation, analogous to (1.8) but using the motor radius r_m, we get:

$$\Delta W = \frac{\tau_L + \tau_R}{r_m} \, \Delta x + \frac{\tau_L - \tau_R}{r_m} \, \Delta y$$

Applying the motor law $\tau = K_m i$, the generalised forces for x and y are:

$$Q_x = \frac{K_m}{r_m} \left(i_L + i_R \right), \qquad Q_y = \frac{K_m}{r_m} \left(i_L - i_R \right) \tag{1.15}$$

Electrical Coordinates

The non-conservative forces in the electrical part are the input voltages and the back-EMF of the motors:

$$\Delta W = \left(u_{m,L} - K_m\dot{\theta}_L\right)\Delta q_L + \left(u_{m,R} - K_m\dot{\theta}_R\right)\Delta q_R$$

Using the same transformation, we get the generalised forces for q_L and q_R:

$$Q_{q_L} = u_{m,L} - \frac{K_m}{r_m}\left(\dot{x}+\dot{y}\right), \qquad Q_{q_R} = u_{m,R} - \frac{K_m}{r_m}\left(\dot{x}-\dot{y}\right) \tag{1.16}$$

Lagrange Equations

Using the fact that this is a linear system and there are no potential energies we get:

$$\mathcal{L} = K - V = \tfrac{1}{2}\,\dot{q}^\top \mathbf{M}\,\dot{q} - V, \qquad V = 0 \quad \Rightarrow \quad \mathcal{L} = \tfrac{1}{2}\,\dot{q}^\top \mathbf{M}\,\dot{q}$$

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}} = \frac{d}{dt}\left(\mathbf{M}\dot{q}\right) = \mathbf{M}\ddot{q}, \qquad \frac{\partial \mathcal{L}}{\partial q} = 0, \qquad \frac{\partial D}{\partial \dot{q}} = \mathbf{N}\dot{q}$$

And finally solving the Euler–Lagrange equation:

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}} - \frac{\partial \mathcal{L}}{\partial q} = -\frac{\partial D}{\partial \dot{q}} + \mathbf{Q}, \qquad \mathbf{Q} = \begin{bmatrix} Q_x \\ Q_y \\ Q_{q_L} \\ Q_{q_R} \end{bmatrix}$$

$$\mathbf{M}\ddot{q} + \mathbf{N}\dot{q} = \mathbf{Q} \tag{1.17}$$

Expanding back out of matrix form we get:

$$\begin{aligned}
\left(m + J_m\frac{2}{r_m^2}\right)\ddot{x} + \left(8\frac{\mu_r}{r_b^2} + 2\mu_{l,x}\right)\dot{x} - \frac{K_m}{r_m}i_L - \frac{K_m}{r_m}i_R &= 0 \\
\left(m + J_m\frac{2}{r_m^2}\right)\ddot{y} + \left(4\frac{\mu_r}{r_b^2} + 2\mu_{l,y}\right)\dot{y} - \frac{K_m}{r_m}i_L + \frac{K_m}{r_m}i_R &= 0 \\
L_m\frac{di_L}{dt} + R_m i_L + \frac{K_m}{r_m}\dot{x} + \frac{K_m}{r_m}\dot{y} - u_{m,L} &= 0 \\
L_m\frac{di_R}{dt} + R_m i_R + \frac{K_m}{r_m}\dot{x} - \frac{K_m}{r_m}\dot{y} - u_{m,R} &= 0
\end{aligned}$$

Or with substitutions, we get the final form of the four differential equations:

$$m_{\text{eq}} = m + \frac{2J_m}{r_m^2}, \qquad b_x = 8\frac{\mu_r}{r_b^2} + 2\mu_{l,x}, \qquad b_y = 4\frac{\mu_r}{r_b^2} + 2\mu_{l,y}$$

$$\boxed{\begin{aligned}
\ddot{x} &= -\frac{b_x}{m_{\text{eq}}}\dot{x} + \frac{K_m}{m_{\text{eq}}r_m}\left(i_L + i_R\right) \\
\ddot{y} &= -\frac{b_y}{m_{\text{eq}}}\dot{y} + \frac{K_m}{m_{\text{eq}}r_m}\left(i_L - i_R\right) \\
\frac{di_L}{dt} &= -\frac{K_m}{r_m L_m}\left(\dot{x}+\dot{y}\right) - \frac{R_m}{L_m}i_L + \frac{u_{m,L}}{L_m} \\
\frac{di_R}{dt} &= -\frac{K_m}{r_m L_m}\left(\dot{x}-\dot{y}\right) - \frac{R_m}{L_m}i_R + \frac{u_{m,R}}{L_m}
\end{aligned}} \tag{1.18}$$

Or in **state-space** form:

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ \dot{x} \\ \dot{y} \\ i_L \\ i_R \end{bmatrix}, \qquad \dot{\mathbf{x}} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \ddot{x} \\ \ddot{y} \\ \frac{di_L}{dt} \\ \frac{di_R}{dt} \end{bmatrix}, \qquad \mathbf{u} = \begin{bmatrix} u_{m,L} \\ u_{m,R} \end{bmatrix}$$

$$\dot{\mathbf{x}} = \mathbf{A}\,\mathbf{x} + \mathbf{B}\,\mathbf{u} \tag{1.19}$$

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{b_x}{m_{\text{eq}}} & 0 & \frac{K_m}{m_{\text{eq}} r_m} & \frac{K_m}{m_{\text{eq}} r_m} \\ 0 & 0 & 0 & -\frac{b_y}{m_{\text{eq}}} & \frac{K_m}{m_{\text{eq}} r_m} & -\frac{K_m}{m_{\text{eq}} r_m} \\ 0 & 0 & -\frac{K_m}{r_m L_m} & -\frac{K_m}{r_m L_m} & -\frac{R_m}{L_m} & 0 \\ 0 & 0 & -\frac{K_m}{r_m L_m} & \frac{K_m}{r_m L_m} & 0 & -\frac{R_m}{L_m} \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{1}{L_m} & 0 \\ 0 & \frac{1}{L_m} \end{bmatrix}$$

1.2.4 Decoupling the x and y Dynamics

To decouple the system, we introduce a set of useful substitutions:

$$\boxed{\begin{array}{llll} i_x = i_L + i_R, & i_y = i_L - i_R, & u_{m,x} = u_{m,L} + u_{m,R}, & u_{m,y} = u_{m,L} - u_{m,R} \\ i_L = \frac{i_x + i_y}{2}, & i_R = \frac{i_x - i_y}{2}, & u_{m,L} = \frac{u_{m,x} + u_{m,y}}{2}, & u_{m,R} = \frac{u_{m,x} - u_{m,y}}{2} \end{array}} \tag{1.20}$$

By substituting our new decoupled currents we get:

$$\begin{aligned} \frac{di_L}{dt} &= \tfrac{1}{2}\frac{di_x}{dt} + \tfrac{1}{2}\frac{di_y}{dt} = -\frac{K_m}{r_m L_m}\dot{x} - \frac{K_m}{r_m L_m}\dot{y} - \frac{R_m}{2L_m} i_x - \frac{R_m}{2L_m} i_y + \frac{1}{2L_m}\left(u_{m,x} + u_{m,y}\right) \\ \frac{di_R}{dt} &= \tfrac{1}{2}\frac{di_x}{dt} - \tfrac{1}{2}\frac{di_y}{dt} = -\frac{K_m}{r_m L_m}\dot{x} + \frac{K_m}{r_m L_m}\dot{y} - \frac{R_m}{2L_m} i_x + \frac{R_m}{2L_m} i_y + \frac{1}{2L_m}\left(u_{m,x} - u_{m,y}\right) \end{aligned}$$

Then by adding and subtracting the two equations to cancel terms we get:

$$\frac{di_x}{dt} = -\frac{2K_m}{r_m L_m}\dot{x} - \frac{R_m}{L_m} i_x + \frac{1}{L_m} u_{m,x}, \qquad \frac{di_y}{dt} = -\frac{2K_m}{r_m L_m}\dot{y} - \frac{R_m}{L_m} i_y + \frac{1}{L_m} u_{m,y} \tag{1.21}$$

And by substituting (1.20) into the first two equations of (1.18):

$$\ddot{x} = -\frac{b_x}{m_{\text{eq}}}\dot{x} + \frac{K_m}{m_{\text{eq}} r_m} i_x, \qquad \ddot{y} = -\frac{b_y}{m_{\text{eq}}}\dot{y} + \frac{K_m}{m_{\text{eq}} r_m} i_y \tag{1.22}$$

And finally:

$$\tilde{\mathbf{x}} = \begin{bmatrix} x \\ \dot{x} \\ i_x \\ y \\ \dot{y} \\ i_y \end{bmatrix}, \qquad \dot{\tilde{\mathbf{x}}} = \begin{bmatrix} \dot{x} \\ \ddot{x} \\ \dfrac{di_x}{dt} \\ \dot{y} \\ \ddot{y} \\ \dfrac{di_y}{dt} \end{bmatrix}, \qquad \tilde{\mathbf{u}} = \begin{bmatrix} u_{m,x} \\ u_{m,y} \end{bmatrix}$$

$$\dot{\tilde{\mathbf{x}}} = \tilde{\mathbf{A}}\,\tilde{\mathbf{x}} + \tilde{\mathbf{B}}\,\tilde{\mathbf{u}} \tag{1.23}$$

$$\tilde{\mathbf{A}} = \left[\begin{array}{ccc|ccc} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & -\dfrac{b_x}{m_{\text{eq}}} & \dfrac{K_m}{m_{\text{eq}} r_m} & 0 & 0 & 0 \\ 0 & -\dfrac{2K_m}{r_m L_m} & -\dfrac{R_m}{L_m} & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & -\dfrac{b_y}{m_{\text{eq}}} & \dfrac{K_m}{m_{\text{eq}} r_m} \\ 0 & 0 & 0 & 0 & -\dfrac{2K_m}{r_m L_m} & -\dfrac{R_m}{L_m} \end{array}\right], \qquad \tilde{\mathbf{B}} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ \dfrac{1}{L_m} & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & \dfrac{1}{L_m} \end{bmatrix}$$

1.3 Drawing Shapes

Rearranging the decoupled equations of motion (1.22), we can express the currents needed to follow a prescribed motion of the tool head:

$$\boxed{i_x = \frac{m_{\text{eq}} r_m}{K_m}\,\ddot{x} + \frac{b_x r_m}{K_m}\,\dot{x}, \qquad i_y = \frac{m_{\text{eq}} r_m}{K_m}\,\ddot{y} + \frac{b_y r_m}{K_m}\,\dot{y}} \tag{1.24}$$

The motor equations for each motor are:

$$u_{m,L} = L_m \frac{di_L}{dt} + R_m i_L + K_m \dot{\theta}_L = L_m \frac{di_L}{dt} + R_m i_L + \frac{K_m}{r_m}(\dot{x} + \dot{y})$$
$$u_{m,R} = L_m \frac{di_R}{dt} + R_m i_R + \frac{K_m}{r_m}(\dot{x} - \dot{y})$$

Adding them, and using (1.20):

$$u_{m,x} = u_{m,L} + u_{m,R} = L_m \left[\frac{di_L}{dt} + \frac{di_R}{dt}\right] + R_m\,[i_L + i_R] + \frac{2K_m}{r_m}\,\dot{x}$$

and similarly by subtracting them, we get the decoupled input voltages, consistent with (1.21):

$$\boxed{u_{m,x} = L_m \frac{di_x}{dt} + R_m i_x + \frac{2K_m}{r_m}\,\dot{x}, \qquad u_{m,y} = L_m \frac{di_y}{dt} + R_m i_y + \frac{2K_m}{r_m}\,\dot{y}} \tag{1.25}$$

The original motor voltages then follow from (1.20):

$$u_{m,L} = \tfrac{1}{2}\left(u_{m,x} + u_{m,y}\right), \qquad u_{m,R} = \tfrac{1}{2}\left(u_{m,x} - u_{m,y}\right) \tag{1.26}$$

Since (1.25) contains $\frac{di_x}{dt}$, and (1.24) contains $\ddot{x}$, the voltages depend on the third derivative of the position. Therefore, for any C^3 functions $x(t)$ and $y(t)$ describing the position of the tool head, we can derive an open-loop analytical solution for the input voltages, even if those voltages are not realistic.

Ellipse

An ellipse with centre (h, k) and semi-axes a and b has the parametric form:

$$x(t) = h + a\cos t, \qquad y(t) = k + b\sin t \tag{1.27}$$

We also need the derivatives up to the third order:

$$\dot{x} = -a\sin t, \quad \ddot{x} = -a\cos t, \quad \dddot{x} = a\sin t, \qquad \dot{y} = b\cos t, \quad \ddot{y} = -b\sin t, \quad \dddot{y} = -b\cos t$$

Substituting into (1.24):

$$\begin{aligned} i_x(t) &= -\frac{m_{\text{eq}} r_m}{K_m}\, a\cos t - \frac{b_x r_m}{K_m}\, a\sin t = -\frac{r_m}{K_m}\, a\left[m_{\text{eq}}\cos t + b_x \sin t\right] \\ i_y(t) &= \frac{r_m}{K_m}\, b\left[b_y \cos t - m_{\text{eq}}\sin t\right] \end{aligned}$$

And into (1.25):

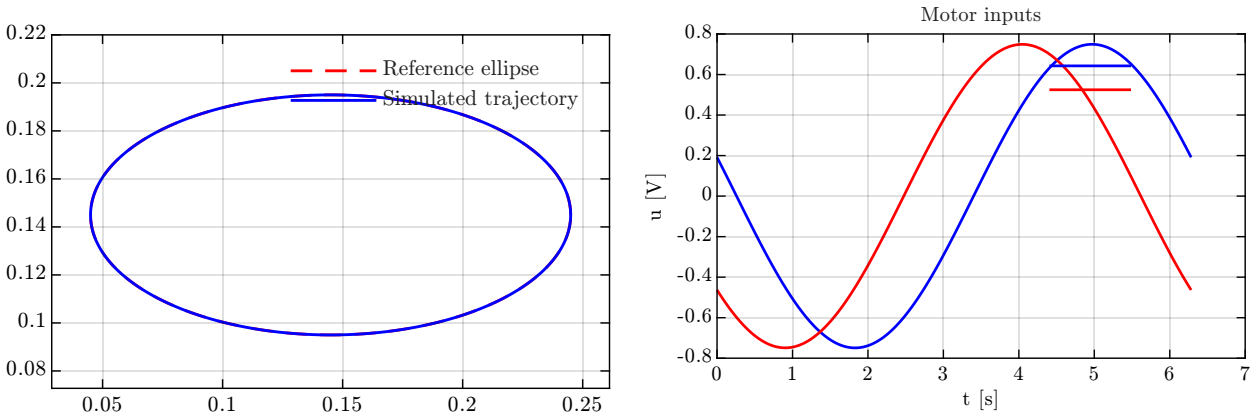
$$\begin{aligned} u_{m,x}(t) &= L_m \frac{r_m}{K_m}\, a\left[m_{\text{eq}}\sin t - b_x\cos t\right] - R_m \frac{r_m}{K_m}\, a\left[m_{\text{eq}}\cos t + b_x \sin t\right] - \frac{2K_m}{r_m}\, a\sin t \\ u_{m,y}(t) &= -L_m \frac{r_m}{K_m}\, b\left[m_{\text{eq}}\cos t + b_y \sin t\right] + R_m \frac{r_m}{K_m}\, b\left[b_y\cos t - m_{\text{eq}}\sin t\right] + \frac{2K_m}{r_m}\, b\cos t \end{aligned}$$

With

$$u_{m,L}(t) = \tfrac{1}{2}\left(u_{m,x}(t) + u_{m,y}(t)\right), \qquad u_{m,R}(t) = \tfrac{1}{2}\left(u_{m,x}(t) - u_{m,y}(t)\right)$$

we obtain the analytical open-loop solution for drawing the ellipse.

Simulating the given system with parameters defined in Table 1, we get near-perfect alignment with the reference ellipse.



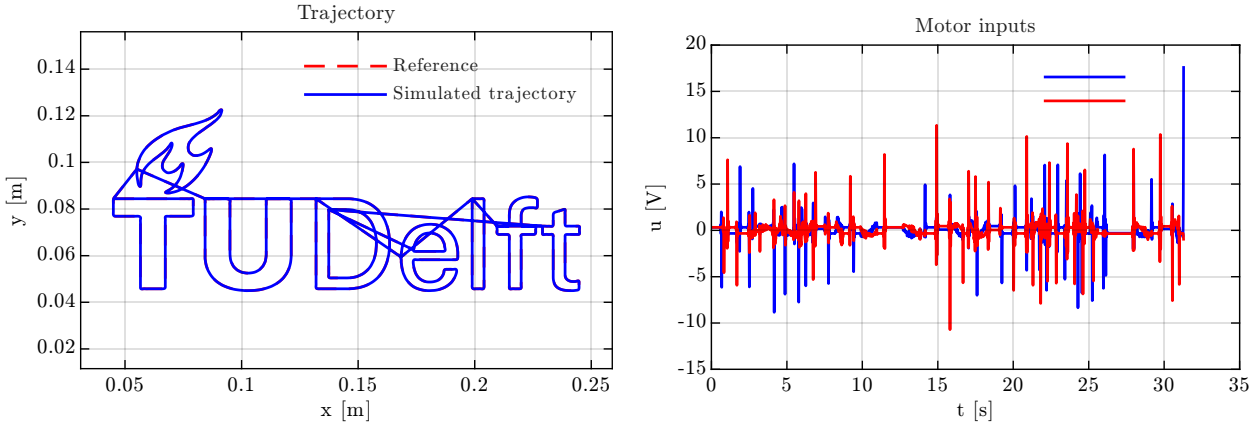
(a) Reference and simulated trajectory.

(b) Open-loop motor voltages.

Figure 2: Open-loop drawing of an ellipse with nominal model parameters.

Arbitrary Drawing

As stated in the derivation, any C^3 function used for $x(t)$ and $y(t)$ will allow us to get an analytical solution. With that in mind we can draw arbitrary shapes by approximating them with quartic splines. Using MATLAB's *bwboundaries* function to detect edges of shapes, in combination with *spapi* which gives us an interpolating quartic spline given a set of points, we can draw the *TU Delft* logo[1].



(a) Reference and simulated trajectory.

(b) Open-loop motor voltages.

Figure 3: Open-loop drawing of the *TU Delft* logo with nominal model parameters.

1.3.1 Modelling the Nozzle

In the following section we model the nozzle of the 3D printer.

The system consists of an electrical and thermodynamical part. A circuit consisting of an inductor L_n, capacitor C_n and a heating resistor R_n. We choose the current through the inductor and the voltage across the capacitor, $i_{L,n}$ and u_c respectively, as first two states of our system. By solving the three Kirchhoff equations we get the following two first-order differential equations:

[1]Different letter boundaries are connected by straight lines so the whole drawing can be one input function

$$L_n \frac{di_{L,n}}{dt} = u_n - u_c, \qquad C_n \frac{du_c}{dt} = i_{L,n} - \frac{u_c}{R_n(T_r)} \tag{1.28}$$

Next, we model the thermal part. Besides the thermal capacitance of the nozzle $C_{t,n}$, given in the task, we also model the thermal capacitance of the resistor C_r[2]. We choose two additional states, temperature of the resistor (T_r) and the temperature of the nozzle (T_n). Losses happening between the resistor and the nozzle are described by $R_{t,n}$, and the heat dissipation to the environment and to the filament is modelled with one variable R_{env}. The connection to the electrical part is the heat dissipated by the resistor $P_r = \frac{u_c^2}{R_n(T_r)}$. This gives us the following two equations which with (1.28) fully describe the system:

$$C_{t,n} \frac{dT_n}{dt} = \frac{T_r - T_n}{R_{t,n}} - \frac{T_n - T_e}{R_{\text{env}}}, \qquad C_r \frac{dT_r}{dt} = P_r - \frac{T_r - T_n}{R_{t,n}} \tag{1.29}$$

with

$$R_n(T_r) = R_0 \left[1 + \alpha \left(T_r - T_0\right)\right] \tag{1.30}$$

where R_0 is the resistance at the reference temperature T_0 and α is the temperature coefficient of resistance.

Due to the square voltage term and the temperature dependence of the resistance the system is very non-linear, although the very different time constants of two systems (thermal states change much slower compared to the electrical) can be exploited. Alternatively, the model can reasonably be simplified by assuming a constant resistance. With that formulation we would get two linear systems connected by a single non-linear term.

1.3.2 Printing a circle of filament

In this section a 1D distributed parameter model for the cooling of a square cross-section filament on a heating plate and a numerical formulation of the problem are derived.

Modelling the system

We take a cross-section of $a \times a$ with length ΔL. Temperature is assumed uniform across the $a \times a$ cross-section to keep the model one dimensional in space. Three faces exchange heat with the environment (T_e), and one face with the heating plate (T_p).

The problem is modelled using the heat equation [2], Fourier's law gives the flux (1.31). Writing the energy density as (1.31) and substituting both into the energy balance yields the heat equation (1.32):

$$\partial_t u = -\nabla \cdot \phi + f, \qquad u = \rho c\, T, \qquad \phi = -\kappa \nabla T \tag{1.31}$$

$$\partial_t T = \frac{\kappa}{\rho c} \nabla^2 T + \frac{f}{\rho c} \tag{1.32}$$

Secondly, we need to describe the heat exchange of the segment ΔL. Using Newton's cooling law we can express the heat transfer through the surfaces as (1.33). Dividing that by V gives us the heat transfer per unit volume (1.34).

[2] If we do not model the thermal capacitance of the resistor, we end up with a differential-algebraic equation (DAE).

$$\Delta A_e = 3a\,\Delta L, \qquad \Delta A_p = a\,\Delta L, \qquad V = a^2\,\Delta L$$

$$\dot{Q} = K_e\,\Delta A_e\,(T_e - T) + K_p\,\Delta A_p\,(T_p - T) \tag{1.33}$$

$$f = \frac{\dot{Q}}{V} = \frac{1}{a}\Big[\,3K_e\,(T_e - T) + K_p\,(T_p - T)\,\Big] \tag{1.34}$$

Substituting everything back into (1.32), gives us the final PDE:

$$\boxed{\partial_t T = k\,\partial_{xx} T + \alpha\,T + \beta} \tag{1.35}$$

$$k = \frac{\kappa}{\rho c}, \qquad \alpha = -\frac{3K_e + K_p}{a\,\rho c}, \qquad \beta = \frac{3K_e T_e + K_p T_p}{a\,\rho c}$$

Front Fixing

Due to needing to solve the problem while the nozzle is still moving, this becomes a moving boundary problem. A method from [1] is used. We introduce a transformation ξ, that maps our space to the interval $[0, 1]$, with $s(t)$ being the position of our boundary, and consequently $\dot{s}(t)$ the speed of the boundary/nozzle.

$$\xi = \frac{x}{s(t)}, \qquad 0 \le \xi \le 1, \qquad s(t) = s_0 + s_1 t, \qquad \dot{s} = s_1 \tag{1.36}$$

Applying a change of variable to our equation (1.35) we get:

$$\frac{\partial T}{\partial x} = \frac{1}{s}\frac{\partial T}{\partial \xi}, \qquad \frac{\partial^2 T}{\partial x^2} = \frac{1}{s^2}\frac{\partial^2 T}{\partial \xi^2}$$

$$\left(\frac{\partial T}{\partial t}\right)_x = \left(\frac{\partial T}{\partial t}\right)_\xi + \frac{\partial T}{\partial \xi}\left(\frac{\partial \xi}{\partial t}\right)_x = \left(\frac{\partial T}{\partial t}\right)_\xi - \frac{x\,\dot{s}}{s^2}\frac{\partial T}{\partial \xi} = \left(\frac{\partial T}{\partial t}\right)_\xi - \frac{\xi\,\dot{s}}{s}\frac{\partial T}{\partial \xi}$$

$$\boxed{\left(\frac{\partial T}{\partial t}\right)_\xi = \frac{\xi\,\dot{s}}{s}\frac{\partial T}{\partial \xi} + \frac{k}{s^2}\frac{\partial^2 T}{\partial \xi^2} + \alpha\,T + \beta} \tag{1.37}$$

Boundary and Initial Conditions

With $T(\xi, t)$ we can place necessary boundary conditions for the problem to be solvable. Firstly, the temperature at the endpoint ($\xi = 1$) at all times equals the temperature of the nozzle which constitutes a Dirichlet condition. Secondly, we assume that the heat dissipation on the ($\xi = 0$) surface is negligible compared to the whole surface, so a Neumann boundary condition is used to model it insulated. Finally, as seen in (1.36), there exists initial length of filament s_0, needed to solve the problem, and its temperature at $t = 0$ is assumed uniform and equal to the nozzle temperature.

$$T(1, t) = T_n, \qquad \partial_\xi T(0, t) = 0, \qquad T(\xi, 0) = T_n, \qquad s_0 > 0 \tag{1.38}$$

Discretisation

The temperature is discretised on a uniform grid in the transformed coordinate $\xi \in [0,1]$, with spacing $\Delta\xi = 1/N$, and in time with step Δt. Time is advanced with an implicit (backward) Euler scheme, the diffusion term $\partial_{\xi\xi}T$ uses a second-order central difference, which is symmetric and produces a tridiagonal matrix and the advection term $\partial_\xi T$ uses a first-order upwind difference.

$$\xi_i = i\,\Delta\xi, \quad i = 0,\dots,N, \qquad t_j = j\,\Delta t, \qquad T_i^j \approx T(\xi_i, t_j)$$

$$\left.\frac{\partial T}{\partial t}\right|_i^j \approx \frac{T_i^j - T_i^{j-1}}{\Delta t}, \qquad \left.\frac{\partial T}{\partial \xi}\right|_i^j \approx \frac{T_{i+1}^j - T_i^j}{\Delta\xi}, \qquad \left.\frac{\partial^2 T}{\partial \xi^2}\right|_i^j \approx \frac{T_{i+1}^j - 2T_i^j + T_{i-1}^j}{\Delta\xi^2} \tag{1.39}$$

$$\frac{T_i^j - T_i^{j-1}}{\Delta t} = \frac{\xi_i\,\dot{s}}{s}\,\frac{T_{i+1}^j - T_i^j}{\Delta\xi} + \frac{k}{s^2}\,\frac{T_{i+1}^j - 2T_i^j + T_{i-1}^j}{\Delta\xi^2} + \alpha\,T_i^j + \beta \tag{1.40}$$

With some substitutions and after inserting the approximations into the transformed PDE gives we get:

$$\gamma_i = \frac{\xi_i\,\dot{s}\,\Delta t}{s\,\Delta\xi} = \frac{i\,\dot{s}\,\Delta t}{s}, \qquad g = \frac{k\,\Delta t}{s^2\,\Delta\xi^2}, \qquad s,\ \dot{s} \text{ evaluated at } t_j$$

$$\boxed{T_i^j\left[\,1 + \gamma_i + 2g - \alpha\,\Delta t\,\right] = g\,T_{i-1}^j + (\gamma_i + g)\,T_{i+1}^j + T_i^{j-1} + \beta\,\Delta t} \tag{1.41}$$

We can see that we arrive at an implicit formulation, where we need to solve a linear system at each time step.

Linear System at $t = t_j$

Writing down a couple of terms we can see that we get a tridiagonal matrix, with the first row representing the Neumann condition and the last the Dirichlet condition.

$$d_i = 1 + \gamma_i + 2g - \alpha\,\Delta t, \qquad e_i = \gamma_i + g, \qquad T_0^j = T_1^j, \qquad T_N^j = T_n$$

$$\begin{bmatrix} 1 & -1 & & & & \\ -g & d_1 & -e_1 & & & \\ & -g & d_2 & -e_2 & & \\ & & \ddots & \ddots & \ddots & \\ & & & -g & d_{N-1} & -e_{N-1} \\ & & & & & 1 \end{bmatrix} \begin{bmatrix} T_0^j \\ T_1^j \\ T_2^j \\ \vdots \\ T_{N-1}^j \\ T_N^j \end{bmatrix} = \begin{bmatrix} 0 \\ T_1^{j-1} + \beta\,\Delta t \\ T_2^{j-1} + \beta\,\Delta t \\ \vdots \\ T_{N-1}^{j-1} + \beta\,\Delta t \\ T_n \end{bmatrix} \tag{1.42}$$

The fact that the matrix is tridiagonal allows us to use specialised solvers like *scipy.linalg.solve_banded*.

After the Boundary Stops (Closed Ring)

After the boundary stops moving, i.e. the circle is closed the matrix needs to be slightly modified, which causes it to lose its tridiagonal form. At the time point of contact the first and last node are connected and become one node, and its temperature is set to the mean of both of them. As the boundary stops moving $\dot{s}$ and consequently γ_i terms disappear.

$$\dot{s} = 0 \;\Rightarrow\; \gamma_i = 0, \qquad s = L, \qquad T_N^j \equiv T_0^j, \qquad g = \frac{k\,\Delta t}{L^2\,\Delta\xi^2}, \qquad d = 1 + 2g - \alpha\,\Delta t$$

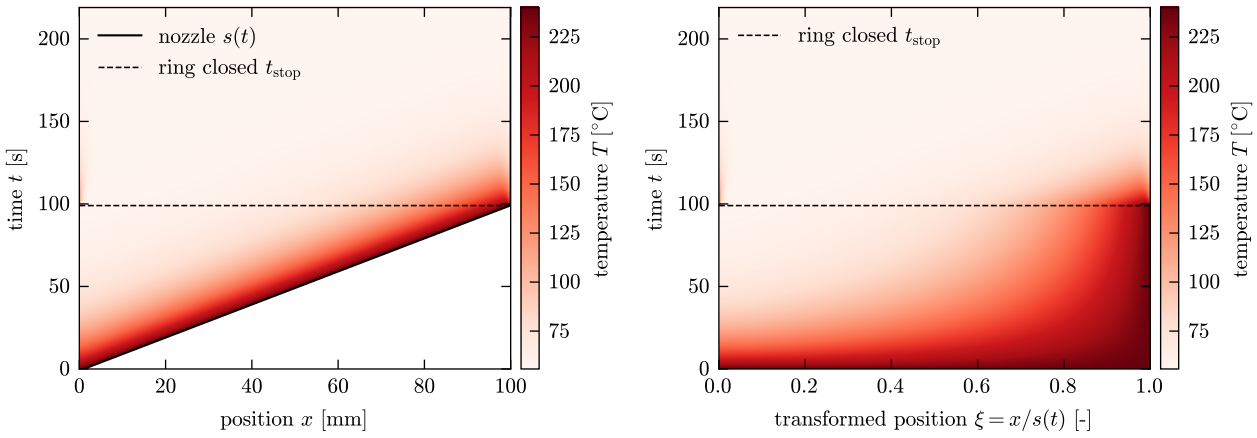
$$\begin{bmatrix} d & -g & & & -g \\ -g & d & -g & & \\ & \ddots & \ddots & \ddots & \\ & & -g & d & -g \\ -g & & & -g & d \end{bmatrix} \begin{bmatrix} T_0^j \\ T_1^j \\ \vdots \\ T_{N-2}^j \\ T_{N-1}^j \end{bmatrix} = \begin{bmatrix} T_0^{j-1} + \beta\,\Delta t \\ T_1^{j-1} + \beta\,\Delta t \\ \vdots \\ T_{N-2}^{j-1} + \beta\,\Delta t \\ T_{N-1}^{j-1} + \beta\,\Delta t \end{bmatrix} \tag{1.43}$$

First step after connection:

$$T_0^{j-1} \leftarrow \tfrac{1}{2}\left(T_0^{j-1} + T_N^{j-1}\right)$$

Simulation

The derived model was simulated in *Python* with the parameters listed in Table 8 (Appendix A).



(a) Physical space x.

(b) Transformed space ξ.

Figure 4: Temperature of the filament while printing and after the ring is closed.

2 Finite State Machine

2.1 Filament Replacement

This section presents the high-level control of the filament replacement, modelled as a Mealy machine since the next state and the output (status message and retry counter) both depend on the current state and input. The machine listens to the external events C (a new colour is requested) and F (the current spool is finished) and to the success or failure feedback of the low-level subsystems. It consists of two coupled sub-machines that are updated together: a *printing* machine (idle, running

or paused) and a *filament replacement* machine that sequences the steps of Q2.1. A failed step is retried up to three times before the replacement is aborted.

2.1.1 Formal Machine Definition

The FSM is defined as the Mealy machine

$$\mathcal{M} = (Q, \Sigma, \Lambda, \delta, \lambda, q_0). \tag{2.1}$$

States. The state $q = (p, f, r)$ of printing state, filament replacement state and retries left,

$$\begin{aligned} Q &= P \times \mathcal{F} \times R, \\ P &= \{\texttt{IDLE}, \texttt{IN_PROG}, \texttt{PAUSED}\}, \\ \mathcal{F} &= \{\texttt{IDLE}, \texttt{MOVE}, \texttt{WIPE}, \texttt{COOL}, \texttt{LOAD}, \texttt{HEAT}\}, \\ R &= \{0, 1, 2, 3\}, \end{aligned} \tag{2.2}$$

where `MOVE` is moving to the wipe location, `WIPE` is wiping the excess filament, `COOL` is the nozzle heat switched off, `LOAD` is loading the new filament and `HEAT` is the nozzle heat switched on. Saving the head location (step 1 of Q2.1) is the action on entering `MOVE`, and resuming the print (step 7) is the action on leaving `HEAT`. The initial state is $q_0 = (\texttt{IDLE}, \texttt{IDLE}, 3)$.

Inputs. At each step the FSM can accept a pair of inputs $\sigma = (e, y)$, where e represents external event i.e C, F or nothing. y represents the feedback of the outcome of the current filament state being executed. It is represented by ok_k if step k succeeded and fail_k if it failed, for each of the five replacement steps $k \in K = \mathcal{F} \setminus \{\texttt{IDLE}\} = \{\texttt{MOVE}, \texttt{WIPE}, \texttt{COOL}, \texttt{LOAD}, \texttt{HEAT}\}$. For example, $\text{fail}_{\texttt{WIPE}}$ means that the wiping did not succeed.

In total there are 3 external events and $1 + 2 \cdot 5 = 11$ feedback values, so the input alphabet has $3 \cdot 11 = 33$ elements:

$$\begin{aligned} E &= \{C,\ F,\ \varepsilon\}, \\ Y &= \{\varepsilon,\ \text{ok}_{\texttt{MOVE}},\ \text{fail}_{\texttt{MOVE}},\ \text{ok}_{\texttt{WIPE}},\ \text{fail}_{\texttt{WIPE}},\ \ldots,\ \text{ok}_{\texttt{HEAT}},\ \text{fail}_{\texttt{HEAT}}\}, \\ \Sigma &= E \times Y. \end{aligned} \tag{2.3}$$

Outputs. The output alphabet λ represents the set of output messages sent to the user on each transition. For example, "filament change in progress", "k successful", "k failure, trying again", "filament change process failed, initiate filament change again" and "filament change already in progress".

Transition and output functions. The transition function $\delta : Q \times \Sigma \to Q$ and output function $\lambda : Q \times \Sigma \to \Lambda$ are given by the transition table and below. Here $\text{next}(k)$ is the step after k in the order $\texttt{MOVE} \to \texttt{WIPE} \to \texttt{COOL} \to \texttt{LOAD} \to \texttt{HEAT}$, and r is the number of retries left, reset to 3 on every new step.

The following cases can occur, depending on the filament replacement state f and the input $\sigma = (e, y)$:

- **Start.** If $f = \texttt{IDLE}$ and the event e is C or F, the replacement starts: the machine moves to `MOVE`, and a running print ($p = \texttt{IN_PROG}$) is paused, while an idle or already paused printer keeps its printing state.

- **Event during a replacement.** If $f \neq$ IDLE and e is C or F, the event is ignored and the state is unchanged; the output message is "filament change already in progress".
- **Step succeeded.** On ok_k in step $f = k$, the machine moves on to next(k), with $r = 3$.
- **Last step succeeded.** On ok_{HEAT} the replacement is finished: the machine returns to $f =$ IDLE, and a paused print is resumed ($p =$ PAUSED $\rightarrow$ IN_PROG).
- **Step failed, retries left.** On fail_k with $r > 0$, the machine stays in step k and the number of retries is reduced by one, $r' = r - 1$, so the step is tried again.
- **Step failed, no retries left.** On fail_k with $r = 0$, the replacement is aborted: the machine returns to $f =$ IDLE with $r = 3$, and the message asks the user to start the filament change again.
- **Anything else.** An input for which the table has no row (e.g. a feedback from a step that is not active) leaves the state unchanged and returns the message NO MATCHING TRANSITION.

Each of these cases has its own output message, which is the output λ of the transition.

Implementation. The machine is implemented as a state transition table loaded into a pandas dataframe in fsm.ipynb. A snapshot of the table is shown in Table 2.

Every row describes one of the transition cases defined above. The first five columns (printing state, filament state, retries left, external input and system feedback) are the conditions that the current state and input must satisfy, and the last four columns are the output state, the retries left and the output message. The entry X represents DON'T CARE condition where the current state of this variable does not influence the output in any manner. An empty input cell means that the signal is absent and must match exactly. The rows are checked from top to bottom and the first row whose conditions all hold is applied; if no row matches, the state is left unchanged and the message NO MATCHING TRANSITION is returned.

The machine is implemented as a state transition table loaded into a pandas dataframe in fsm.ipynb. A snapshot of the table is shown in Table 2.

Table 2: Snapshot of the state transition table (state names abbreviated, X is a DON'T CARE, empty cells mean no signal).

p	f	e	y	r	p'	f'	message	r'
IDLE	IDLE	C		X	IDLE	MOVE	filament change in progress	3
IN_PROG	IDLE	C		X	PAUSED	MOVE	filament change in progress	3
X	MOVE		ok_MOVE	X	X	WIPE	moving to wiping location successful	3
X	MOVE		fail_MOVE	3	X	MOVE	failure, trying again	2
X	MOVE		fail_MOVE	0	X	IDLE	process failed, initiate again	3
PAUSED	HEAT		ok_HEAT	X	IN_PROG	IDLE	nozzle heating successful	3
X	WIPE	C	X	X	X	WIPE	filament change already in progress	X

The model was checked against a set of different test cases (test_cases.csv).

2.1.2 Transition

The two sub-machines are shown in Figures 5 and 6. Transitions are labelled as *input* [*guard condition*] / *action*, where r is the number of retries left. A C or F event starts the replacement only when the filament replacement machine is in its idle state; while a replacement is running, further C or F events are ignored (self-loop with the message "filament change already in progress"). If a

C or F event arrives in the same step as a feedback signal, the external event takes precedence and the feedback is discarded.

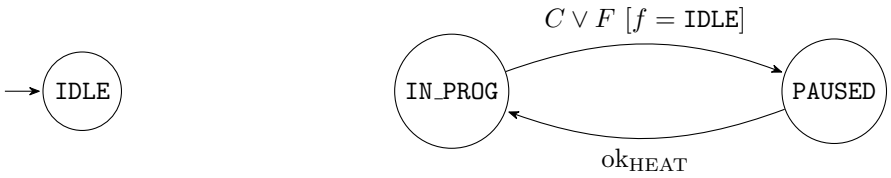


Figure 5: Printing sub-machine with state $p \in P$. The transition from `IDLE` to `IN_PROG` (print started) belongs to the print job handling and is not modelled here.

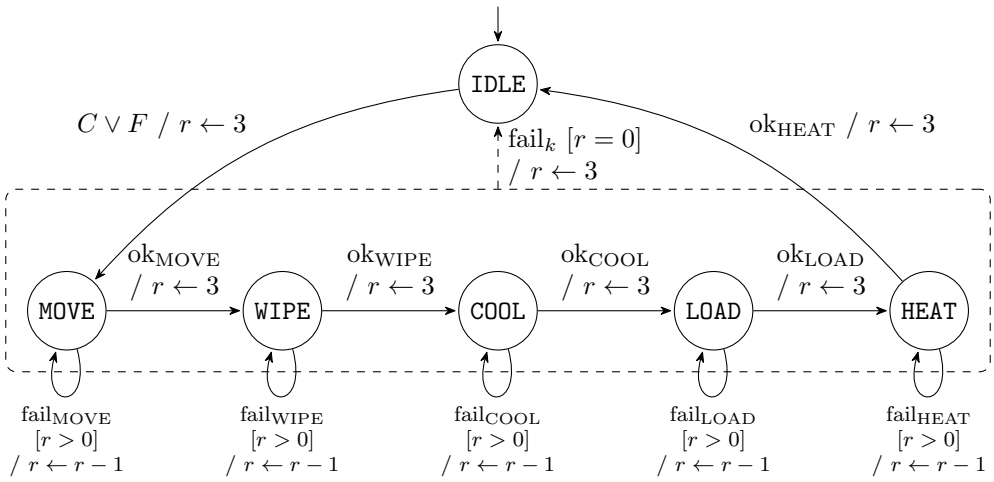


Figure 6: Filament replacement sub-machine with state $f \in \mathcal{F}$. For readability the five $\text{fail}_k\,[r = 0]$ transitions from the steps k back to `IDLE` are drawn as a single dashed arrow from the box around the steps. The two sub-machines are synchronised: the $C \vee F$ event and the ok_{HEAT} event act on both machines in the same step.

2.2 Parked Detection

The head is parked at the origin when both the horizontal and the vertical motion have hit their end of line, i.e. a motion was commanded but the corresponding motor did not rotate. The detector is the Deterministic Finite Automaton (DFA)

$$\mathcal{A} = (Q_A, \Sigma_A, \delta_A, q_{A,0}, F_A) \tag{2.4}$$

with

$$\begin{aligned} Q_A &= \{q_0, q_1, q_2, q_3\}, \\ \Sigma_A &= M \times m = \{(H,+), (H,0), (V,+), (V,0)\}, \\ q_{A,0} &= q_0, \qquad F_A = \{q_3\}, \end{aligned} \tag{2.5}$$

where the states have the meaning

- q_0: no end of line detected,
- q_1: horizontal end of line (left side) detected,

- q_2: vertical end of line (bottom side) detected,
- q_3: both detected, the head is parked.

The transition function $\delta_A : Q_A \times \Sigma_A \to Q_A$ is given in Table 3 and drawn in Figure 7.

Table 3: Transition function δ_A of the parked detector.

q	$(H,+)$	$(H,0)$	$(V,+)$	$(V,0)$
$\to q_0$	q_0	q_1	q_0	q_2
q_1	q_0	q_1	q_1	q_3
q_2	q_2	q_3	q_0	q_2
$* q_3$	q_2	q_3	q_1	q_3

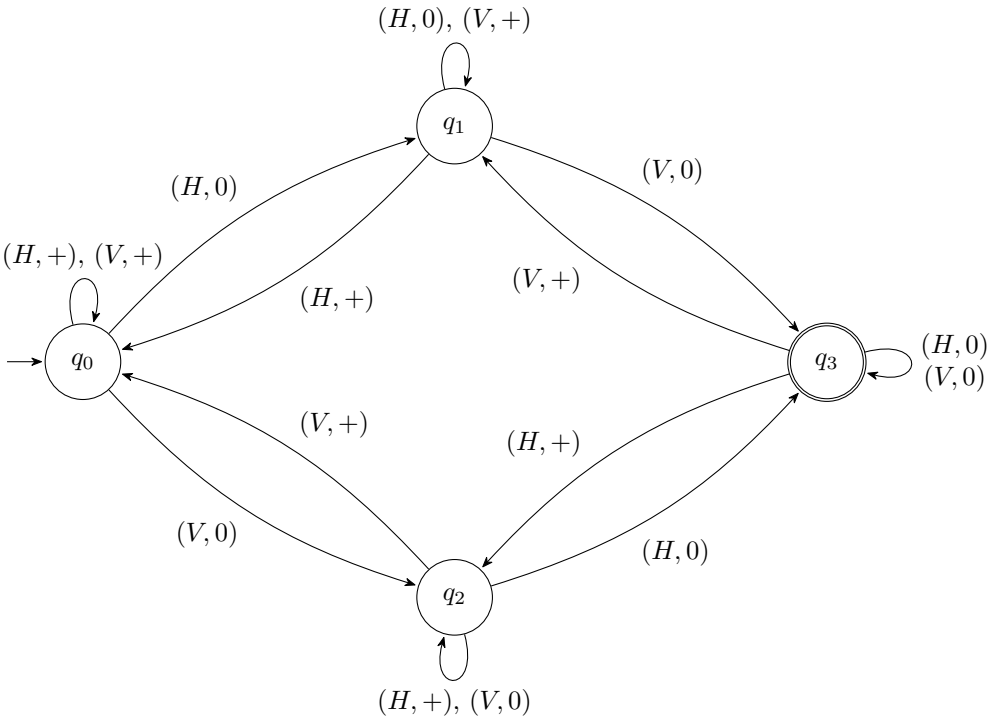


Figure 7: Transition of the parked detector $\mathcal{A}$. The accepted language is the set of sequences σ that end in q_3.

2.3 Detecting the Other Corners

The automaton is extended to the motions H_x, H_y and V and to the angular velocities $+$, $-$ and 0,

$$\begin{aligned} \mathcal{A}' &= (Q_A, \Sigma'_A, \delta'_A, q_0, F_A), \\ \Sigma'_A &= \{H_x, H_y, V\} \times \{+, -, 0\}, \end{aligned} \tag{2.6}$$

with Q_A, q_0 and F_A as before. The transition function $\delta'_A : Q_A \times \Sigma'_A \to Q_A$ is given in Table 4 and drawn in Figure 8.

Table 4: Transition function δ'_A of the corner detector.

q	$(H_x, \pm)$	$(H_x, 0)$	$(H_y, \pm)$	$(H_y, 0)$	$(V, \cdot)$
$\rightarrow q_0$	q_0	q_1	q_0	q_2	q_0
q_1	q_0	q_1	q_1	q_3	q_1
q_2	q_2	q_3	q_0	q_2	q_2
$*\, q_3$	q_2	q_3	q_1	q_3	q_3

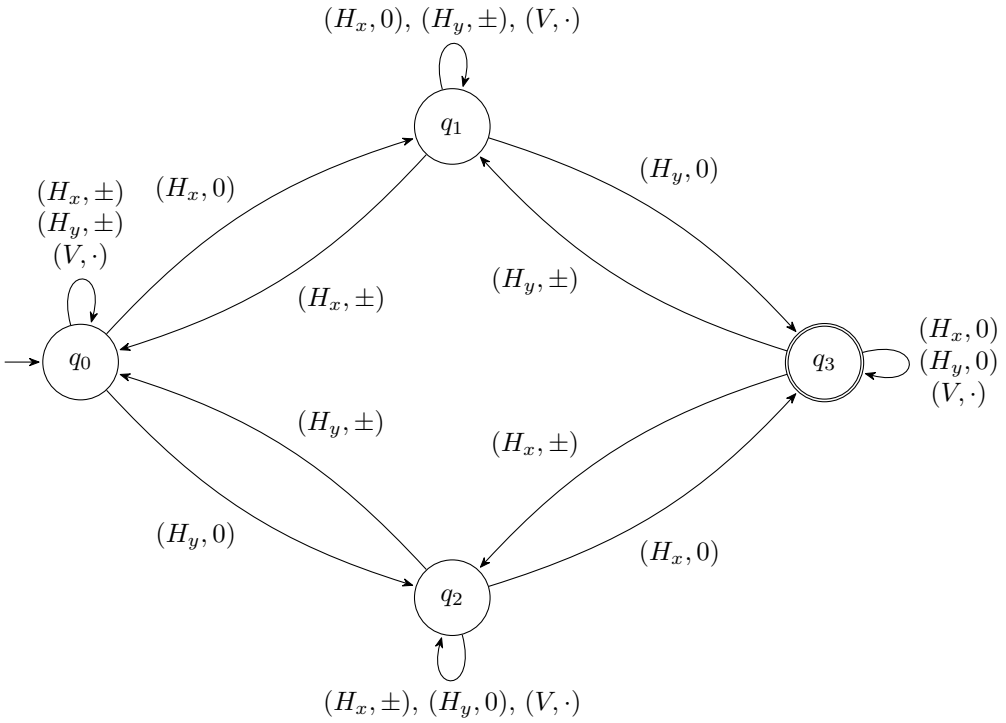


Figure 8: Transition of the corner detector $\mathcal{A}'$.

3 Hybrid System

3.1 Part 1: Formal Definition

We define the hybrid state machine as

$$H = (Q, X, U, Y, f, h, Init, E, G, R).$$

It combines three smaller machines running in parallel: the nozzle machine $\mathcal{N}$, the printer machine $\mathcal{P}$ and the filament machine $\mathcal{M}$.

In mode q the continuous state flows as $\dot{x} = f(q, x, u)$. An edge is taken as soon as its guard is true, and z^+ denotes the value of z right after the transition. There are no separate invariants.

Q: set of modes. One mode from each machine, $Q = Q_{\mathcal{N}} \times Q_{\mathcal{P}} \times Q_{\mathcal{M}}$:

$$\begin{aligned} Q_{\mathcal{N}} &= \{\text{COOL}, \text{HEATING_UP}, \text{HEATED}, \text{COOLING}\}, \\ Q_{\mathcal{P}} &= \{\text{IDLE}, \text{WAIT_HOT}, \text{TRAVEL_TO_START}, \text{PRINT_PATH}, \\ &\qquad \text{PAUSED}, \text{COMPLETED}\}, \\ Q_{\mathcal{M}} &= \{\text{IDLE}, \text{MOVING_TO_WIPE}, \text{WIPING}, \text{NOZZLE_COOLING}, \\ &\qquad \text{LOADING_FILAMENT}, \text{NOZZLE_HEATING}, \text{READY_TO_RESUME}\}. \end{aligned}$$

WAIT_HOT, TRAVEL_TO_START and PRINT_PATH together form the mode IN_PROGRESS of $\mathcal{P}$.

X: continuous state. The printer state $x_p = (x, y, \dot{x}, \dot{y}, T_n, T_r)$ (head position and velocity, nozzle and heater temperatures) plus the variables of Table 5. We use the reduced plant, where the fast electrical dynamics are instantaneous ($L_m \approx 0$, $u_c = u_n$).

Table 5: Variables in X besides the printer state x_p.

Symbol	Meaning
L	Length of filament used so far
τ	Time spent in the current mode; a timer with $\dot{\tau} = 1$ that is reset to 0 on every edge
k	Index of the current point in the path, which is an array of coordinates that the printer follows; $k \in \{1, \ldots, n\}$
(x_s, y_s)	Coordinates of the head saved when the printer is paused
req, *heated*	Request to the nozzle to start heating; indicator that the nozzle is heated
paused	Indicator that the printer is in the paused state
change_active, *change_done*	Indicator that a filament change is in progress; indicator that a filament change has finished

$U = U_D \times U_C$: inputs. $U_D = \{C, START\}$ are the user events (colour change, start). The continuous inputs U_C are listed in Table 6. They are fixed by the mode, as given in Section 3.2. The motors are driven by $\mathcal{M}$ when it is not IDLE and by $\mathcal{P}$ otherwise.

Table 6: Continuous inputs U_C.

Symbol	Code variable	Meaning
(u_{mL}, u_{mR}, u_n)	`mode_motor`, `mode_nozzle`	Left and right motor voltages and heater voltage; each is either off or on, depending on the mode
Extruder switch	`is_extruding`	Whether the extruder is on
$r = (x_c, y_c)$	`command_x`, `command_y`	Commanded head position that the motors drive towards

Y: outputs. The outputs are the head position (x, y), the nozzle temperature T_n, the length of filament used L, and the current mode of each machine $(q_{\mathcal{N}}, q_{\mathcal{P}}, q_{\mathcal{M}})$. Here the mode of a machine is the step it is currently in, for example HEATED for $\mathcal{N}$, PAUSED for $\mathcal{P}$ or WIPING for $\mathcal{M}$.

f: flow map. The reduced CoreXY plant (`sys_reduced`): a damped head mass driven by the two coupled motors, and a two-node nozzle thermal model heated with power $u_n^2/R_n(T_r)$. The other variables follow Table 5.

h: output map. h computes the outputs from the current mode $q = (q_{\mathcal{N}}, q_{\mathcal{P}}, q_{\mathcal{M}})$ and continuous state x, so $y_{out} = h(q, x)$. It does not change the values; it only selects which quantities are observed:

$$y_{out} = h(q, x) = \big(\, x,\ y,\ T_n,\ L,\ q_{\mathcal{N}},\ q_{\mathcal{P}},\ q_{\mathcal{M}} \,\big).$$

The first four entries are the head position (x, y), T_n and L, read directly from x_p and L. The last three come from q. The remaining variables ($\dot{x}$, $\dot{y}$, T_r, τ, k, the saved coordinates and the flags) are internal and not part of the output.

$Init$: initial state. $Init = \{(q_0, x_0)\}$ contains a single element:

$$\begin{aligned} q_0 &= (\text{COOL}, \text{IDLE}, \text{IDLE}), \\ x_p(0) &= (x, y, \dot{x}, \dot{y}, T_n, T_r) = (0, 0, 0, 0, T_e, T_e), \\ (L, \tau, k) &= (0, 0, 1), \\ (\mathit{req}, \mathit{heated}, \mathit{paused}, \mathit{change_active}, \mathit{change_done}) &= (\text{false}, \text{false}, \text{false}, \text{false}, \text{false}), \end{aligned}$$

where T_e is the ambient temperature.

Shorthand for the guards. The conditions in Table 7 are used to keep the guards short.

Table 7: Shorthand used in the guards.

Symbol	Meaning
arr	Head has arrived at its target: $\lvert x - x_c\rvert < \varepsilon$ and $\lvert y - y_c\rvert < \varepsilon$, with $\varepsilon = 0.5\,\text{mm}$
F	Spool is empty: $L \geq L_{spool}$
T_{hi}	Temperature at which the nozzle counts as hot, $T_{hi} = 350\,\text{K}$
$\mathit{arr}_{>0.5}$	arr has held for more than 0.5 s
$(T_n \geq T_{hi})_{>1}$	$T_n \geq T_{hi}$ has held for more than 1 s
$C^{\uparrow}$	Button C has just been pressed (rising edge, $0 \to 1$)

E: edges. $E \subseteq Q \times Q$ is the set of allowed transitions $e = (q, q')$. It is the union of the edges of the three machines, which are listed in Section 3.2:

$$\begin{aligned} E &= E_{\mathcal{N}} \cup E_{\mathcal{P}} \cup E_{\mathcal{M}}, \\ e &= (q, q'), \quad \text{e.g. } e = (\text{COOL}, \text{HEATING_UP}) \in E_{\mathcal{N}}. \end{aligned}$$

G: guards. $G(e)$ is the condition on x and u under which the edge e is taken:

$$\begin{aligned} G(\text{COOL}, \text{HEATING_UP}) &= \mathit{req}, \\ G(\text{HEATING_UP}, \text{HEATED}) &= (T_n \geq T_{hi})_{>1}. \end{aligned}$$

The edges IN_PROGRESS $\to$ PAUSED in $\mathcal{P}$ and IDLE $\to$ MOVING_TO_WIPE in $\mathcal{M}$ share the guard $C^{\uparrow} \vee F$, so they fire together.

R: resets. $R(e)$ gives the values z^+ of the variables right after e is taken:

$$R(\text{IDLE}, \text{WAIT_HOT}): \ req^+ = \text{true},$$
$$R(\text{PRINT_PATH}, \text{PRINT_PATH}): \ k^+ = k + 1.$$

Every edge also resets $\tau^+ = 0$ (not repeated in the tables), and any variable not listed keeps its value, $z^+ = z$.

3.2 Part 2: The Machines

Each machine is shown as a flowchart that gives its modes, the inputs u used in each mode, and the edges with their guard G and reset R, followed by a short explanation.

3.2.1 Nozzle Machine $\mathcal{N}$

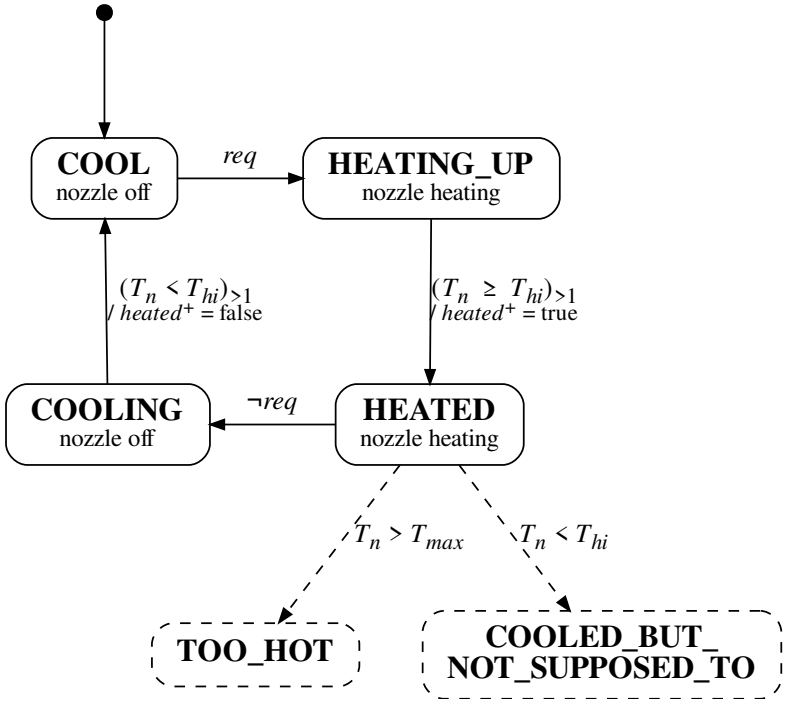


Figure 9: Nozzle machine, in the notation of this report. The dashed modes are failure modes of the implementation, which are not part of H. Edge labels show the guard G, and below it the reset R after the slash. Inputs u are given in each mode.

- The nozzle starts in COOL with the heater off. When heating is requested (*req*), it moves to HEATING_UP and heats.
- After $T_n \geq T_{hi}$ has held for more than 1 s it is HEATED and *heated* becomes true. The machine keeps heating and does not regulate T_n further.
- When *req* is withdrawn the heater is switched off (COOLING), and after $T_n < T_{hi}$ for more than 1 s it returns to COOL with *heated* false.

3.2.2 Printer Machine $\mathcal{P}$

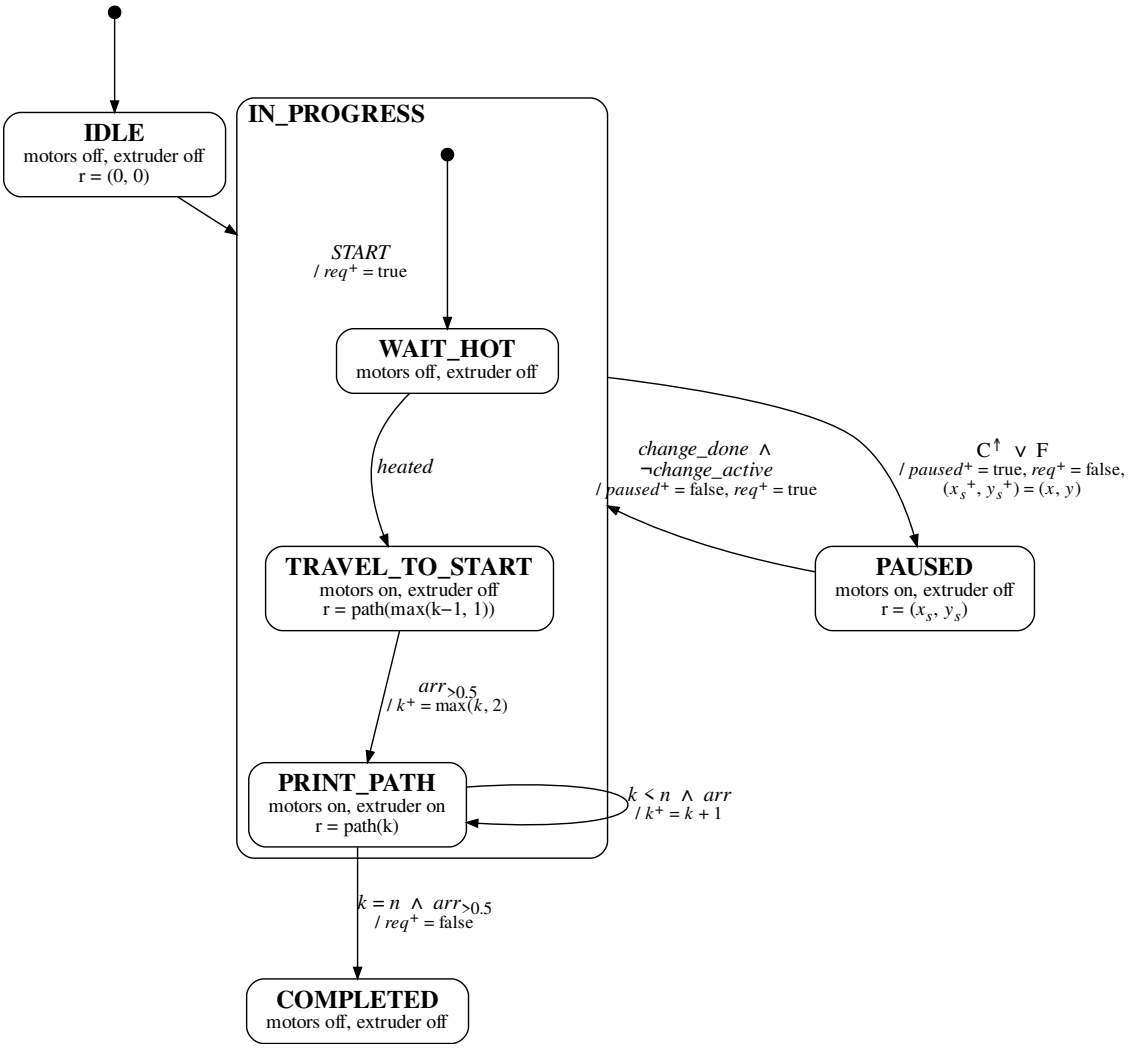


Figure 10: Printer machine, in the notation of this report. Edge labels show the guard G, and below it the reset R after the slash. Inputs u are given in each mode.

- *START* requests heating and the printer waits in WAIT_HOT until the nozzle is *heated*.
- The head then travels to the start of the path and prints it point by point, increasing k each time it arrives. After the last point the printer is COMPLETED.
- On $C^{\uparrow}$ or F the printer pauses from any sub-mode of IN_PROGRESS and saves the head position (x_s, y_s). When the change is done it re-enters WAIT_HOT, so it waits for the nozzle, travels back to path$(k-1)$ and continues from point k.

3.2.3 Filament Machine $\mathcal{M}$

The wipe location is (x_w, y_w), the wipe time is t_w and the load time is t_l.

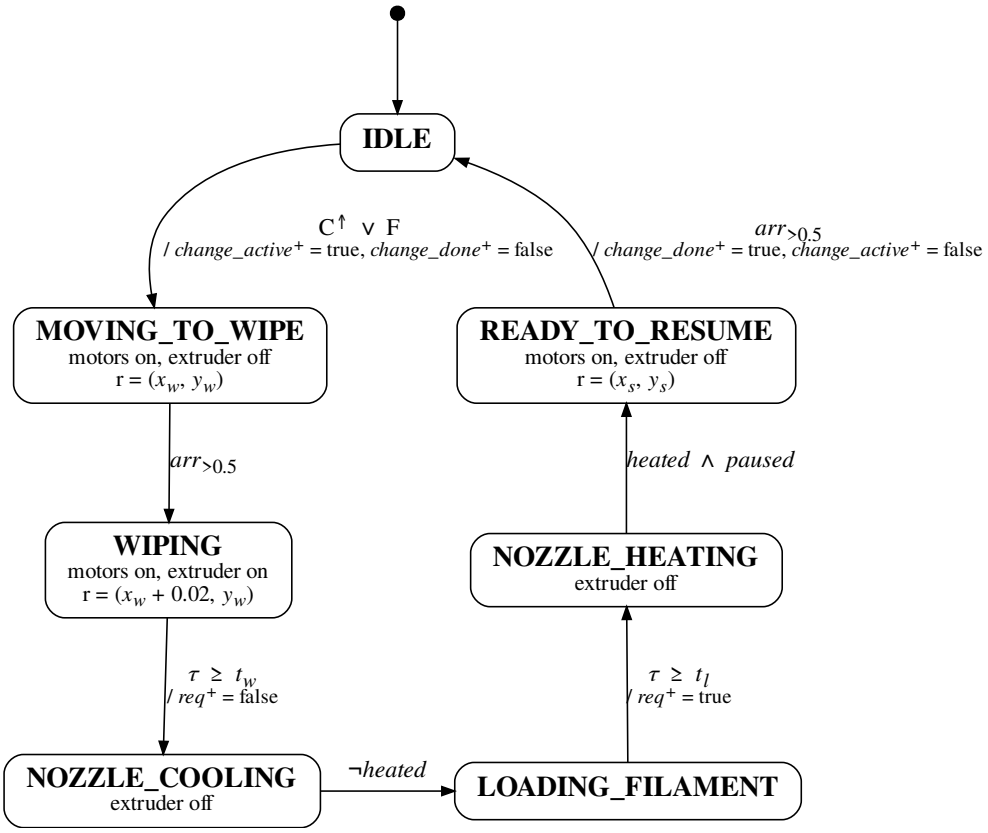


Figure 11: Filament machine, in the notation of this report. Edge labels show the guard G, and below it the reset R after the slash. Inputs u are given in each mode.

- On $C^{\uparrow}$ or F the change starts: the head moves to the wipe location (x_w, y_w) and wipes with the extruder on for $t_w = 3\,\mathrm{s}$.
- Heating is then switched off (*req* false). Once the nozzle has cooled ($\neg\, heated$), new filament is loaded for $t_l = 5\,\mathrm{s}$.
- Heating is requested again. When the nozzle is *heated* and the printer is paused, the head returns to (x_s, y_s) and *change_done* becomes true, which releases the printer.

Note. The nozzle flowchart shows two failure modes of the implementation (dashed). The implementation also has a *FAILED* mode and a manual pause and resume input. The definitions above ignore these failure modes and all retries, and cover only the successful filament change.

3.3 Part 3: Drawing a Square

Path Tracking The square is drawn from a list of points $\mathrm{path}(k) = (x_k, y_k)$, $k = 1, \ldots, n$, with $n = 25$.

- The path runs counter-clockwise through the corners $(0.05, 0.05)$, $(0.25, 0.05)$, $(0.25, 0.25)$, $(0.05, 0.25)$ (in metres) and back to the start.

- Each side has five intermediate points, so consecutive points are 33.3 mm apart.
- In PRINT_PATH the head is commanded to path(k) with the extruder on.
- Once *arr* holds, the index is increased to $k + 1$ and the head moves to the next point.
- The square is therefore drawn point by point, until $k = n$.

Cases. The same square is drawn twice with a filament change in the middle of the print:

1. C is triggered half-way through the print;
2. F happens: the spool is empty half-way through the print.

In both cases the head is moved to the wipe location $(0, 0.5)$ m, and the part printed before the change is shown in red and the part after it in green.

3.3.1 Case 1: C Triggered

C is pressed at $t \approx 69.6$ s, half-way through the print (after two sides of the square). The printer pauses, the head wipes at $(0, 0.5)$ m, the filament is loaded and the nozzle reheats, and printing resumes at $t \approx 96.0$ s (red to green). The square is complete at $t \approx 163$ s.

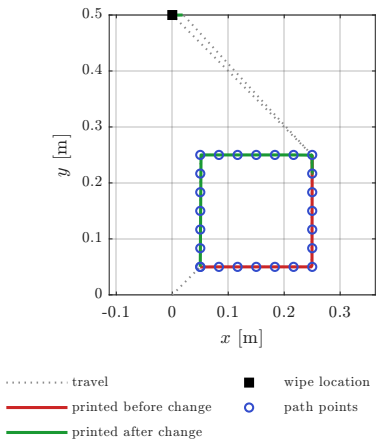


Figure 12: Case 1, head path: printed before the change (red), after (green), and travel moves.

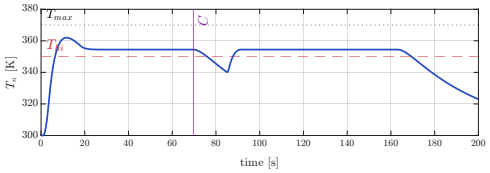


Figure 13: Case 1, nozzle temperature T_n with the thresholds T_{hi} and T_{max}.

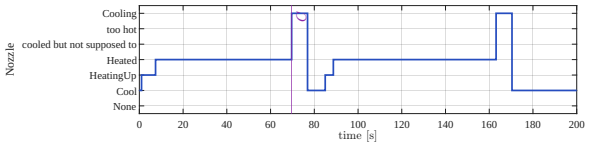


Figure 14: Case 1, nozzle machine mode.

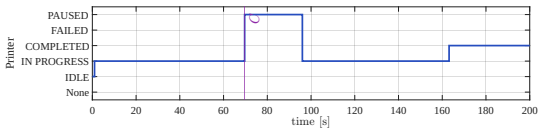


Figure 15: Case 1, printer machine mode.

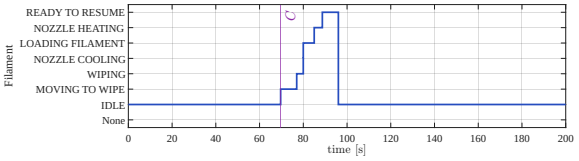


Figure 16: Case 1, filament machine mode.

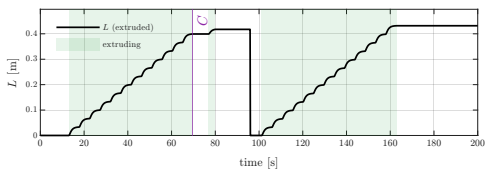


Figure 17: Case 1, extruded filament length L against L_{spool}.

3.3.2 Case 2: F Triggered

C is never pressed. The spool holds half of the filament needed for the square, so F occurs at $t \approx 69.6\,\text{s}$ and the same change sequence runs, resuming at $t \approx 96.0\,\text{s}$. The replacement spool has the same length, so F occurs a second time at $t \approx 158\,\text{s}$ (resuming at 185.6 s) and the square is complete at $t \approx 191\,\text{s}$.

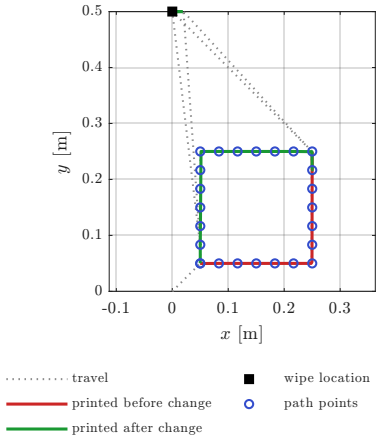


Figure 18: Case 2, head path: printed before the change (red), after (green), and travel moves.

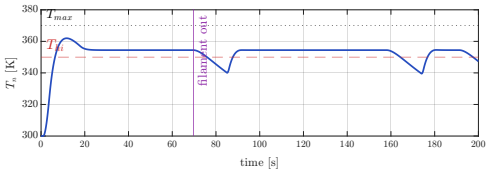


Figure 19: Case 2, nozzle temperature T_n with the thresholds T_{hi} and T_{max}.

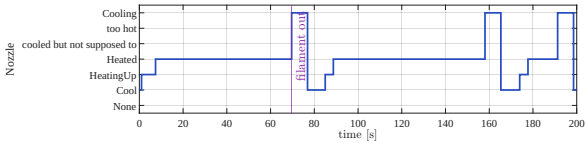


Figure 20: Case 2, nozzle machine mode.

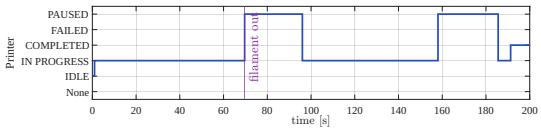


Figure 21: Case 2, printer machine mode.

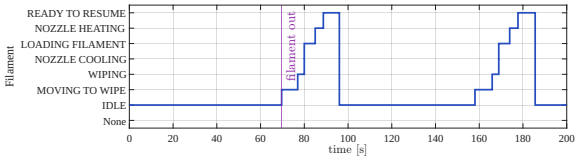


Figure 22: Case 2, filament machine mode.

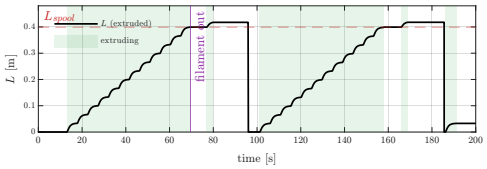


Figure 23: Case 2, extruded filament length L against L_{spool}.

References

[1] John Crank. *Free and Moving Boundary Problems*. Clarendon Press, Oxford, 1984.

[2] Ton J. J. van den Boom, Manuel Mazo Jr., and Riccardo M. G. Ferrari. Modeling of dynamical systems. Lecture notes for the course SC42155 (draft), Delft Center for Systems and Control, Delft University of Technology, September 2026.

A Filament Simulation Parameters

Table 8: Parameters of the filament cooling simulation.

Description	Symbol	Value
Density	ρ	$1240\,\mathrm{kg/m^3}$
Specific heat	c	$1800\,\mathrm{J/(kg\,K)}$
Thermal conductivity	κ	$0.13\,\mathrm{W/(m\,K)}$
Cross-section side	a	$2\,\mathrm{mm}$
Heat transfer coefficient, environment	K_e	$10\,\mathrm{W/(m^2\,K)}$
Heat transfer coefficient, heating plate	K_p	$200\,\mathrm{W/(m^2\,K)}$
Environmental temperature	T_e	$26.85\,^\circ\mathrm{C}$
Heating plate temperature	T_p	$60\,^\circ\mathrm{C}$
Initial filament length	s_0	$1\,\mathrm{mm}$
Nozzle speed	s_1	$1\,\mathrm{mm/s}$
Ring length	L	$100\,\mathrm{mm}$
Nozzle temperature	T_n	$240\,^\circ\mathrm{C}$
Number of intervals	N	200
Time step	Δt	$0.05\,\mathrm{s}$
Simulated time after closing	–	$120\,\mathrm{s}$